

Classification Regression

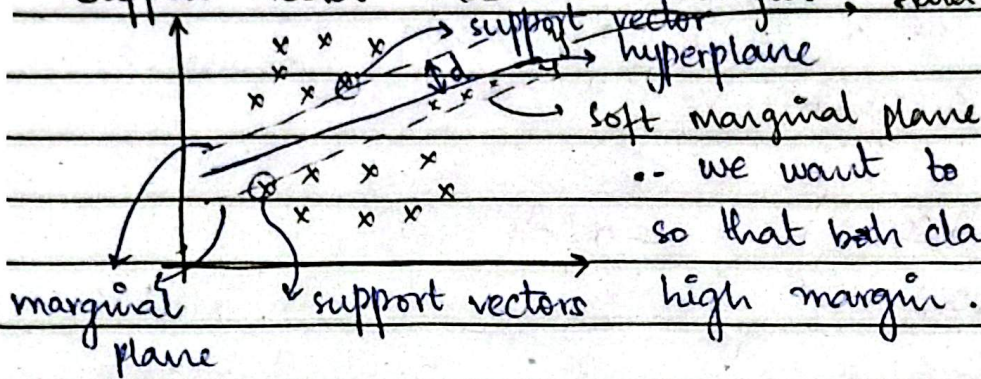
- SVM Classifier
- SVM Regressor

SVM (Support Vector Machine)

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

Support Vector Machine Classifier, Hard marginal plane.



we want to maximize distance d so that both classes are separable by high margin.

Eqⁿ of straight line :-

$$y = mx + c \text{ or } y = \theta_0 + \theta_1 x \text{ or } y = \beta_0 + \beta_1 x$$

$$\text{or } y = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

$$\text{or } y = w^T x + b$$

$$y = mx + c$$

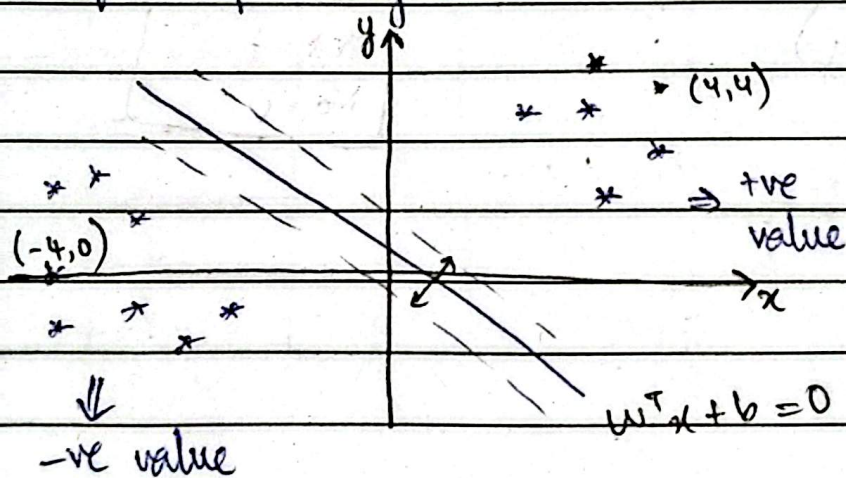
both are same

$$ax + by + c = 0$$

$$by = -ax - c$$

$$y = -\frac{a}{b}x - \frac{c}{b}, \quad m = -\frac{a}{b}, \quad c = -\frac{c}{b}$$

Eqⁿ of straight line can also be written as :- $ax + by + c = 0$.



$$3x + 2y + 4 = 0$$

$$3(-4) + 2(0) + 4$$

$$-12 + 0 + 4$$

$$= -8 \Rightarrow -ve$$

$$3(4) + 2(4) + 4 = 0$$

$$12 + 8 + 4$$

$$24 \Rightarrow +ve$$

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

$$w^T x_1 + b = +1$$

$$w^T x_2 + b = -1$$

 $w \rightarrow$ magnitude

 $\rightarrow$ vector ($\vec{w}$) direction

$$\vec{w} \left\{ \frac{w^T (x_1 - x_2)}{\|w\|} = 2 \right\} \text{ Maximize "d" between these marginal plane.}$$

$$\text{Maximize}_{(w,b)} = \frac{2}{\|w\|} \Rightarrow \text{Marginal plane difference "d".}$$

Constraint:-

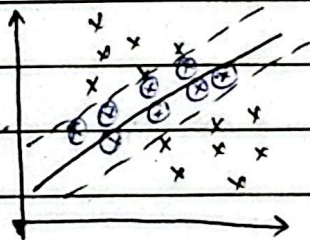
$$\text{such that } y_i \begin{cases} +1 & w^T x + b \geq 1 \\ -1 & w^T x - b \leq -1 \end{cases}$$

For all accurate data points:-

$$\boxed{y * (w^T x_i + b) \geq 1} \Rightarrow \text{constraint}$$

$$\text{Maximize}_{(w,b)} = \frac{2}{\|w\|} \Rightarrow \text{Minimize}_{(w,b)} = \frac{\|w\|}{2}$$

$$\text{Cost function:- Minimize}_{(w,b)} \frac{\|w\|}{2} + \sum_{i=1}^n c_i \xi_i \quad \left\{ \begin{array}{l} \text{Summation of the} \\ \text{distance of misclassification} \\ \text{points from the} \\ \text{marginal plane.} \end{array} \right.$$

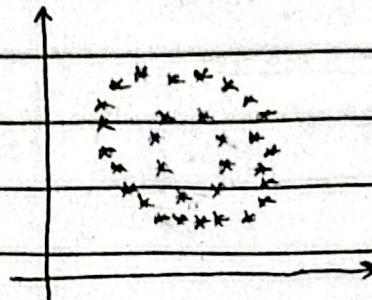
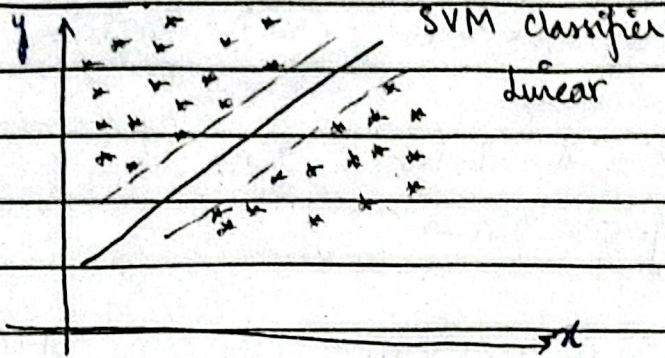


⊙ misclassified examples
c=8

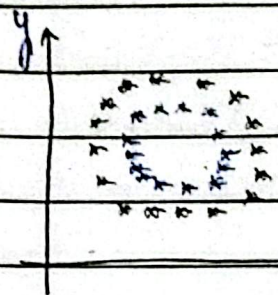
Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

SVM Kernels



Data points Not linearly separable

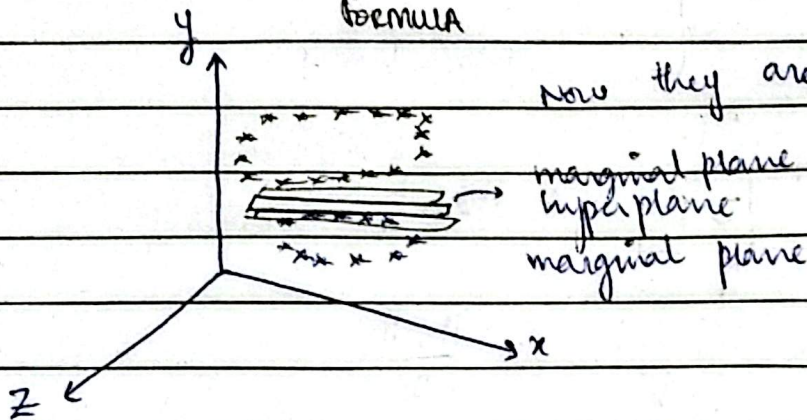


⇒ TRANSFORMATION

MATHEMATICAL

Formula

Need for SVM kernel



Eg:-

Not linearly separable

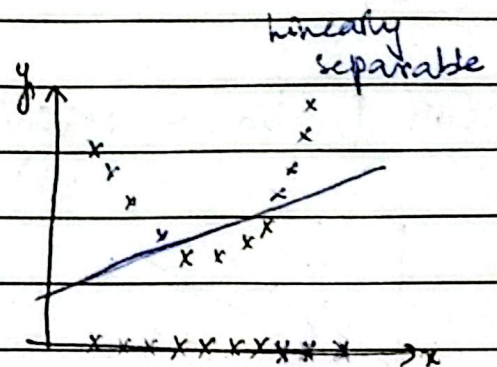
1d

⇒

TRANSFORMATION

$$y = x^2$$

Kernel



① Polynomial Kernel

② RBF Kernel

③ Sigmoid Kernel

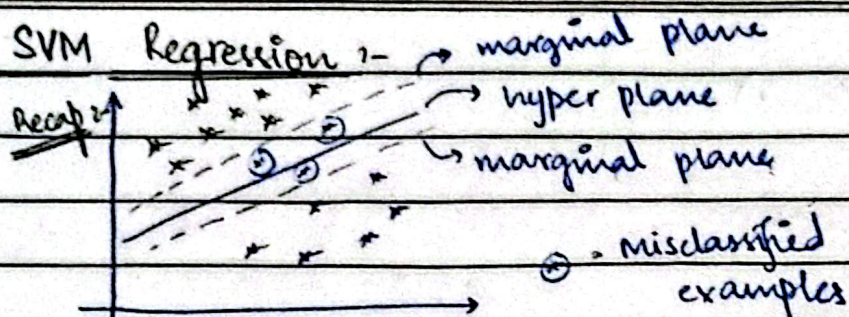
Page No. _____

LOS®

Signature _____

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---



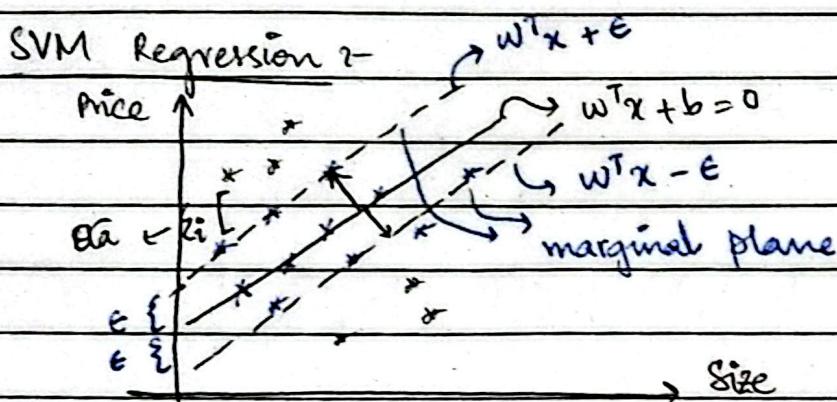
Cost func :- Minimize $\frac{\|w\|}{2} + C \sum_{i=1}^n \xi_i$

w, b

of misclassified examples.

Hyperparameter

$y * (w^T x_i + b) \geq 1 \Rightarrow \text{Constraint}$



Cost func :-

Minimize $\frac{\|w\|}{2}$

(w, b)

Constraint :-

$|y_i - w^T x_i| \leq \epsilon \rightarrow \text{It is very good}$

$|y_i - \hat{y}_i| \leq \epsilon$

Cost func :-

Minimize $\frac{\|w\|}{2} + C \sum_{i=1}^n |\xi_i|$

(w, b)

→ Hinge loss

Hyperparameter

Constraint :-

$|y_i - w^T x_i| \leq \epsilon + |\xi_i|$

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

SVM Numerical

(label +1) :-

$(3, 1), (3, -1), (6, 1), (6, -1)$

(label -1) :-

$(1, 0), (0, 1), (0, -1), (-1, 0)$

three support vectors

$$s_1 = (1, 0) \rightarrow -1$$

$$s_2 = (3, 1) \rightarrow +1$$

$$s_3 = (3, -1) \rightarrow +1$$

Augment vectors with a 1 to include bias b as part of the weight vector.

$$\tilde{s}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \tilde{s}_2 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \quad \tilde{s}_3 = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

$$\tilde{w} = \begin{pmatrix} w \\ b \end{pmatrix}$$

$$w^T x + b = \tilde{w}^T \tilde{x}$$

$$\sum_i \alpha_i (\tilde{s}_i \cdot \tilde{s}_k) = y_k$$

$$\tilde{s}_1 = (1, 0, 1)^T, \quad \tilde{s}_2 = (3, 1, 1)^T, \quad \tilde{s}_3 = (3, -1, 1)^T$$

$$\tilde{s}_1 \cdot \tilde{s}_1 = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 1 + 0 + 1 = 2$$

$$\tilde{s}_2 \cdot \tilde{s}_1 = 3 \cdot 1 + 1 \cdot 0 + 1 \cdot 1 = 3 + 0 + 1 = 4$$

$$\tilde{s}_3 \cdot \tilde{s}_1 = 3 \cdot 1 + (-1) \cdot 0 + 1 \cdot 1 = 3 + 0 + 1 = 4$$

$$\tilde{s}_2 \cdot \tilde{s}_2 = 3^2 + 1^2 + 1^2 = 11$$

$$\tilde{s}_2 \cdot \tilde{s}_3 = 3 \cdot 3 + 1 \cdot (-1) + 1 \cdot 1 = 9 - 1 + 1 = 9$$

$$\tilde{s}_3 \cdot \tilde{s}_3 = 3^2 + (-1)^2 + 1^2 = 11$$

$$\sum_i \alpha_i (\tilde{s}_i \cdot \tilde{s}_k) = y_k$$

For $k=1$:-

$$\alpha_1 (\tilde{s}_1 \cdot \tilde{s}_1) + \alpha_2 (\tilde{s}_2 \cdot \tilde{s}_1) + \alpha_3 (\tilde{s}_3 \cdot \tilde{s}_1) = y_1 = -1$$

$$2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1$$

For $k=2$:-

$$\alpha_1 (\tilde{s}_1 \cdot \tilde{s}_2) + \alpha_2 (\tilde{s}_2 \cdot \tilde{s}_2) + \alpha_3 (\tilde{s}_3 \cdot \tilde{s}_2) = y_2 = +1$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = +1$$

For $k=3$:-

$$\alpha_1 (\tilde{s}_1 \cdot \tilde{s}_3) + \alpha_2 (\tilde{s}_2 \cdot \tilde{s}_3) + \alpha_3 (\tilde{s}_3 \cdot \tilde{s}_3) = y_3 = +1$$

$$4\alpha_1 + 9\alpha_2 + 11\alpha_3 = +1$$

3x3 linear system

$$2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = +1$$

$$4\alpha_1 + 9\alpha_2 + 11\alpha_3 = +1$$

$$\begin{bmatrix} 2 & 4 & 4 \\ 4 & 11 & 9 \\ 4 & 9 & 11 \end{bmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} -1 \\ +1 \\ +1 \end{pmatrix}$$

$$\begin{bmatrix} 5/2 & -1/2 & -1/2 \\ -1/2 & 3/8 & -1/8 \\ -1/2 & -1/8 & 3/8 \end{bmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} -7/2 \\ 3/4 \\ 3/4 \end{pmatrix}$$

Take its inverse

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} -1 \\ +1 \\ +1 \end{pmatrix} \left(\begin{bmatrix} 2 & 4 & 4 \\ 4 & 11 & 9 \\ 4 & 9 & 11 \end{bmatrix} \right)^{-1}$$

Inverse

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} -3/2 \\ 3/4 \\ 3/4 \end{pmatrix}$$

$$\alpha_1 = -3.5, \alpha_2 = 0.75, \alpha_3 = 0.75$$

Date: _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

$$\tilde{w} = \sum_{i=1}^3 \alpha_i \tilde{s}_i$$

$$\tilde{w} = \alpha_1 \tilde{s}_1 + \alpha_2 \tilde{s}_2 + \alpha_3 \tilde{s}_3 = -3.5 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

$$\tilde{w} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

$$w = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, b = -2$$

$$f(x) = \text{sign}(w^T x + b) = \text{sign}(x_1 - 2)$$

$$x_1 - 2 = 0 \quad \text{decision boundary}$$

$$x_1 = 2$$

Verify margin inequalities

$$y_k (w^T s_k + b) = 1$$

Report: Using SVM, Linear Regression, & Logistic Regression for Regression Problems.

SVM $\rightarrow$ SVR Support Vector Regression

• SVR extends SVM to continuous outputs by solving a convex optimization that fits a function within an ϵ -insensitive tube - errors smaller than ϵ are ignored, errors larger than ϵ are penalized. The result depends only on support vectors (data near tube edges).

• Kernel trick :- SVR uses the same kernels (linear, RBF, polynomial) to learn non-linear regression functions by working in feature space.

• Variants :- Common variants are ϵ -SVR and ν -SVR; hyperparameters C , ϵ (or ν), and kernel parameters control bias-variance & tube width.

Practical strengths :- good in high-dimensional settings, robust to outliers near the tube, & can model non linear relations with kernels.

Logistic Regression

Logistic Regression models the probability of a binary outcome using a log-odds (logit) link - it's a generalized linear model (GLM) for binomial data. It's primarily a classification / probability model, not a standard continuous regression.

How to use it for regression :-

For continuous outcomes bounded in $(0, 1)$ like a rate, use a transformation (e.g. $\logit(y)$ & then linear regression) or use specialized models (Beta regression). For ordered categorical

outcomes use ordinal logistic regression.

Research Article

↳ Using LR, SVM and a hybrid model.

SVM

↳ $f(x) = w^T x + b$

LR

↳ $y(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}}$

Hybrid Model

↳ Improved Accuracy, lower MSE, RMSE.

Multiple Linear Regression

significant predictors

↳ Number of daily positive cases.

↳ Number of cured cases.

↳ Number of deceased cases.